

Unified FWA Field Framework: Gravity and Baryon Junction Dynamics

1. Fundamental Position of FWA

Fractal-Wave Analysis (FWA) is formulated as a single field framework governed by the fundamental equation:

$$E_{FWA}[\Psi] = 0$$

where Ψ denotes the relevant wave configuration and E_{FWA} is a nonlinear operator whose projections describe physical phenomena across widely separated scales. Two of these projections are of particular interest:

- **the gravitational sector**, responsible for galactic dynamics, and
- **the junction sector**, describing coherent three-branch baryon structures observed experimentally.

The central claim is not that FWA replaces GR or QCD, but that both phenomena can be represented within a single wave-operator framework whose quantitative consequences can be tested and potentially falsified.

2. FWA Gravity Sector

2.1 Newtonian Limit and Operator Modification

In the Newtonian limit of General Relativity, the gravitational potential $\Phi(r)$ satisfies:

$$\nabla^2 \Phi = 4\pi G \rho_b$$

which yields the familiar Keplerian fall-off:

$$v^2(r) = GM / r$$

Thus GR predicts $v(r) \propto r^{-1/2}$ for large radii. However, observations of spiral galaxies show flat rotation curves:

$$v(r) \rightarrow v_0 \quad (r \rightarrow \infty)$$

which implies:

$$d\Phi / dr = v_0^2 / r \Rightarrow \Phi(r) \sim v_0^2 \ln r$$

This logarithmic potential is incompatible with GR + baryons alone. FWA resolves this by modifying the operator rather than adding a new source:

$$\nabla^2 \Phi - 4\pi G \rho_b - \Delta_{FWA}[\Phi] = 0$$

where Δ_{FWA} is the asymptotic wave-operator contribution. Its large-scale behavior directly yields $\Phi(r) \sim v_0^2 \ln r$ without introducing dark matter.

2.2 Effective Density Interpretation

Substituting the logarithmic potential into the Poisson operator gives:

$$\nabla^2 \Phi \sim v_0^2 / r^2$$

which corresponds to an effective density:

$$\rho_{eff}(r) = v_0^2 / (4\pi G r^2)$$

In GR this term is interpreted as a dark-matter halo. In FWA it arises naturally from the modified operator Δ_{FWA} , not from an additional physical source.

3. FWA Junction Sector

3.1 Coherent Three-Branch Wave State

A baryon junction is represented as a coherent three-branch wave configuration:

Ψ_J = Σ A_k e^{iϕ_k}, ϕ_k = 2πk/3 + δϕ_k

The unperturbed phases satisfy the root-of-unity cancellation:

Σ e^{i 2πk / 3} = 0

which FWA interprets as a dynamic-zero condition of a perfectly balanced junction.

3.2 Symmetry-Error Functional

Departures from perfect symmetry are quantified by the functional:

Q_J^2 = |Σ A_k e^{iϕ_k}|^2

For equal amplitudes A_k = 1 and small distortions δϕ_k, the leading contribution is quadratic:

Q_J^2 ∝ |Σ δϕ_k e^{i 2πk / 3}|^2

3.3 Dynamical Recursion and Transport

The junction evolves under the general FWA recursion:

Ψ_{n+1} = F(Ψ_n, V_n, d_n, T_n)

Linearization around a stationary junction yields Ψ_{n+1} ≈ λ_J Ψ_n. Over a rapidity interval Δy, this leads to the transport law:

dN_B / dy = N_0 exp[-λ_J(y - y_0)]

4. Unified Structure of the FWA Framework

Both sectors are projections of the same fundamental operator E_FWA[Ψ] = 0:

- Galactic scales: E_grav[Φ, ρ_b] = 0
- Hadronic scales: E_junction[Ψ_J, T] = 0

5. Final Consolidated Block (Book-Ready)

Unified FWA Field Equation

E_FWA[Ψ] = 0

Gravity Projection

∇²Φ - 4πGρ_b - Δ_FWA[Φ] = 0
Φ(r) ~ v_0^2 ln r, v(r) ~ v_0

Junction Projection

Ψ_J = Σ A_k e^{iϕ_k} | Q_J^2 = |Σ A_k e^{iϕ_k}|^2
Ψ_{n+1} ≈ λ_J Ψ_n | dN_B/dy = N_0 e^{-λ_J(y-y_0)}